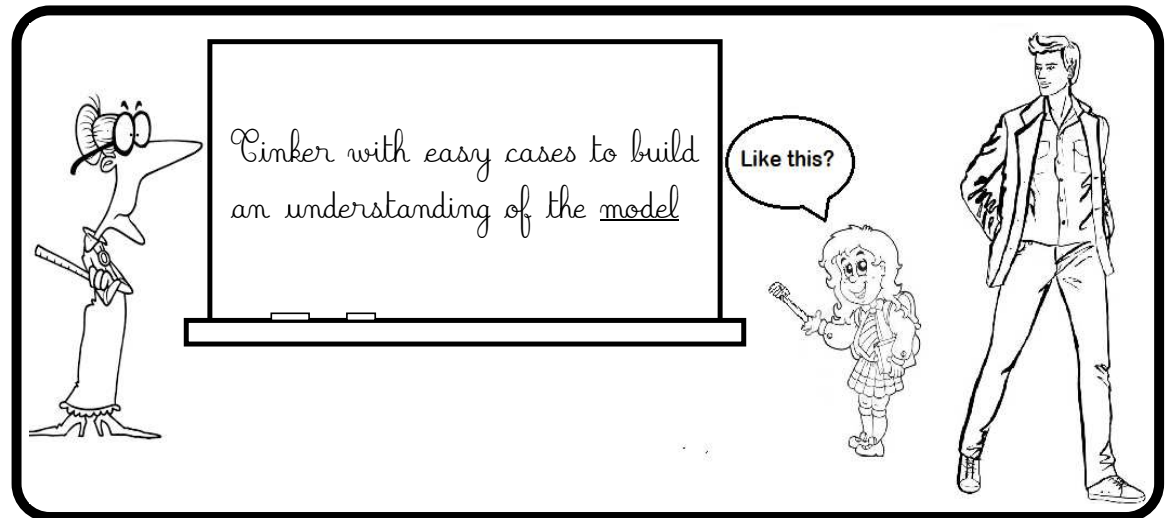
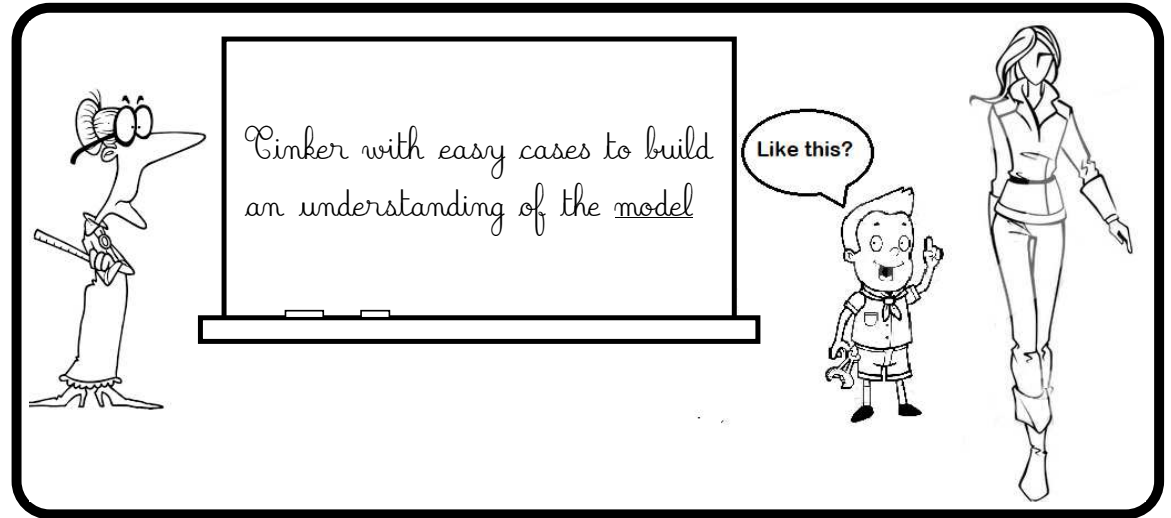


Foundations of Computer Science

Lecture 2

Discrete Objects and Proof

The Cast of Discrete Objects
Some Basic Proofs



A taste of discrete math and computing (ebola, speed dating, friendship networks)

\$100 <i>Distinct</i> subsets with the same sum.	\$1,000 Domino Program	\$10 Create the best ‘math’-cartoon.
5719825393567961346558155629 5487945882843158696672157984 4767766531754254874224257763 1855924359757732125866239784 4289776424589197647513647977 7967131961768854889594217186 2572967277666133789225764888 1294587141921952639693619381 4764413635323911361699183586 1474343641823476922667154474 2578649763684913163429325833 5161596985226568681977938754 2242632698981685551523361879 7474189614567412367516833398 6211855673345949471748161445 494271623349872129251848674 5516264359672753836539861178 5854762719618549417768925747 5313691171963952518124735471 6737691754241231469753717635 4292388614454146728246198812 4468463715866746258976552344 2638621731822362373162811879 1258922263729296589785418839 4482279727264797827654899397 8749855322285371162986411895 1116599457961971796683936952 3879213273596322735993329751 9212359131574159657168196759 3351223183818712673691977472 8855835322812512868896449976 4332859486871255922555418653 2428751582371964453381751663 6738481866868951787884276161 8794353172213177612939776215 2989694245827479769152313629 6117454427987751131467589412 2761854485919763568442339436 6884214746997985976433695787 8671829218381757417536862814 9431156837244768326468938597 4788448664674885883585184169 3624757247737414772711372622 9361819764286243182121963365 9893315516156422581529354454 5913625989853975289562158982 8313891548569672814692858479 2265865138518379114874613969 3477184288963424358211752214 6321349612522496241515883378 1796439694824213266958886393 636625253175995567694496585 8545458545636898974365938274 3362291186211522318566852576 846447386637547496734772855 2892857564355262219965984217 4296693937661266715382241936 8634764617265724716389775433 8415234243182787534123894858 2267353254454872616182242154 4689911847578741473186337883 4428766787964834371794565542 7146295186764167268433238125 2273823813572968577469388278 6686132721336864457635223349 3161518296576488158997146221 1917611425739928285147758625 3516431537343387135357237754 7549684656732941456945632221 2397876675349971994958579984 4675844257857378792991889317 2832515241382937498614676246 8755442772953263299368382378 9833662825734624455736638328 5298671253425423454611152788 9857512879181186421823417538 1471226144331341144787865593 3545439374321661651385735599 6735367616915626462272211264 2141665754145475249654938214 8481747257332513758286947416 9961217236253576952797397966 9941237996445827218665222824 6242177493463484861915865966 4344843511782912875843632652 7568842562748136518615117797 2776621559882146125114473423 6174299197447843873145457215 5387584131525787615617563371 5317693353372572284588242963 6612142515552593663955966562 1314928587713292493616625427 2446827667287451685939173534 9786693878731984534924558138 2926718838742634774778713813 37914262744975966411969142899 2831727715176299968774951996 3281287353463725292271916883 9954744594922386766735519674 3414339143545324298853248718	d_1 0 100 d_2 01 00 d_3 110 11 $d_3d_1d_3 =$ 110 0 110 11 100 11 $\rightarrow$ 1100110 1110011 Goal: Want same top and bottom. Domino program: Input: dominos Output: sequence that works or say it can't be done	Create a cartoon to illustrate/make fun of some discrete math you learned in this class.

Today: Discrete Objects and Proof

1 Discrete Objects

- Sets
- Sequences
- Graphs

2 Proof

- In 4 rounds of the speed-dating app, no one meets more than 12 people.
- x^2 is even “is the same as” x is even
- Among *any* 6 people is a 3-clique or 3-war.
- **Axioms.** The Well-Ordering Principle.
- $\sqrt{2}$ is not rational.

- ① Collection of objects, order does not matter: $F = \{f, o, x\}$; $V = \{a, e, i, o, u\}$.
 $F \cap V = \{o\}$ $F \cup V = \{a, e, f, i, o, u, x\}$ $\overline{F} = ?$

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What is “...?”

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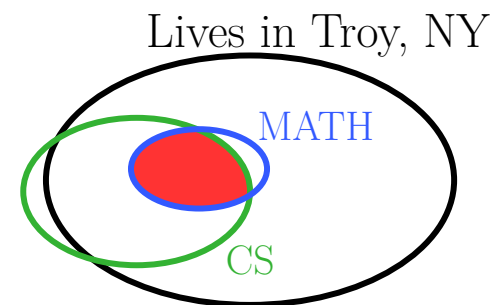
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- ⑨ Venn Diagrams are a convenient way to represent sets.



- ① List of objects: order and repetition matter.

$$tap \neq taap \neq atp$$

Sequences

- 1 List of objects: order and repetition matter.

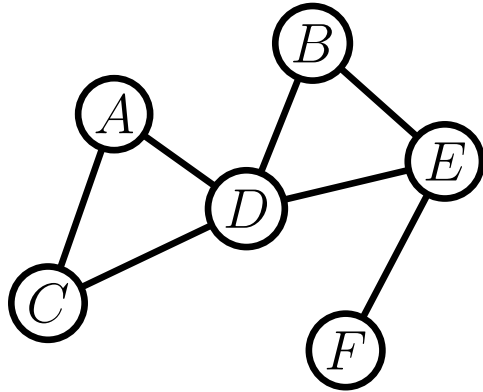
$$tap \neq taap \neq atp$$

- 2 We are mostly concerned with *binary sequences* composed of *bits* (**ASCII** code).

t	a	p
01110100	01100001	01110000

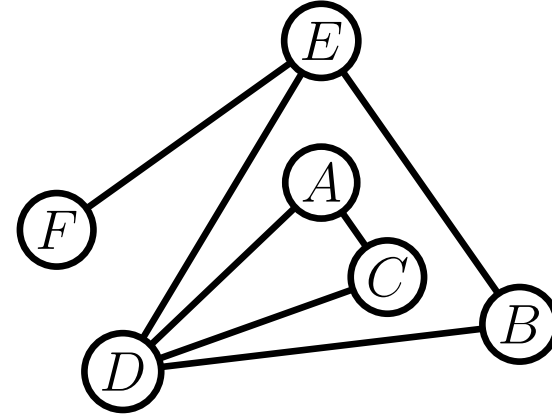
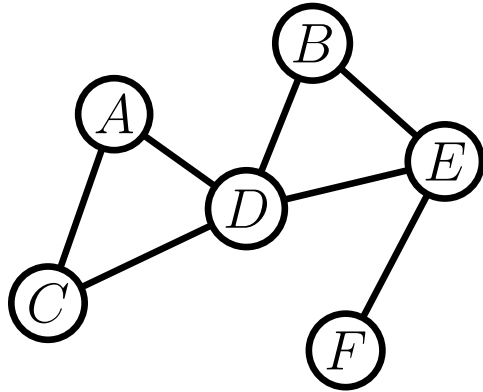
Graphs

Friendships between Alice, Bob, Charles, David, Edward, Fiona:



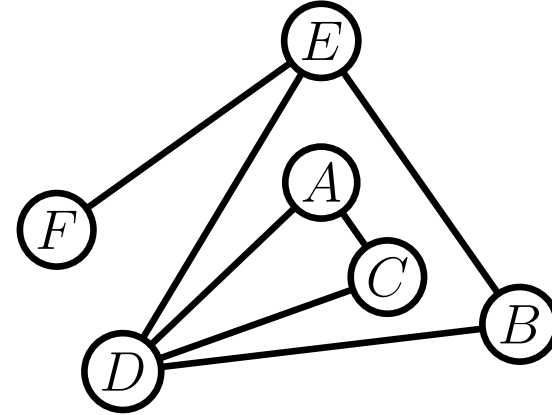
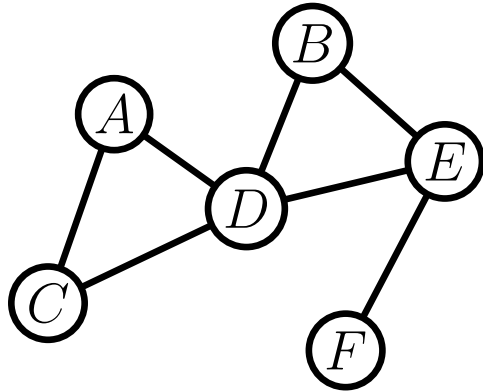
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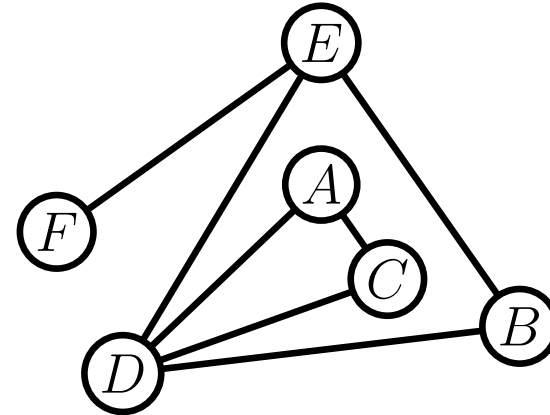
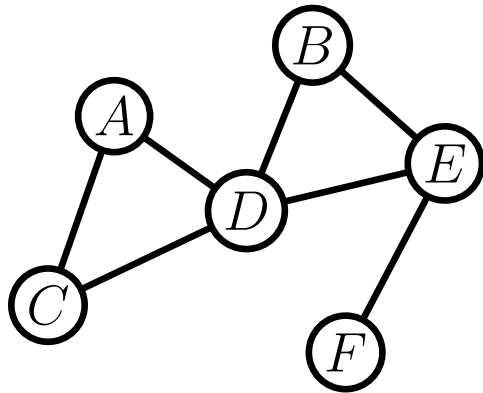
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Graphs

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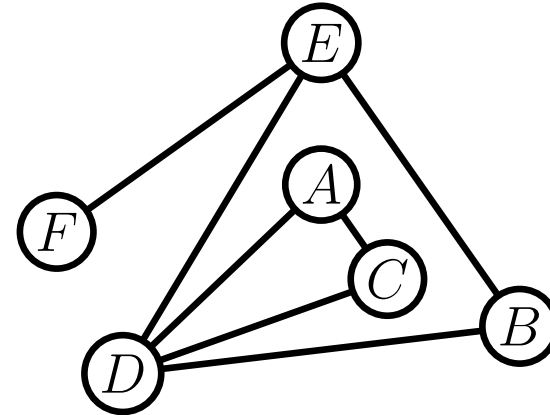
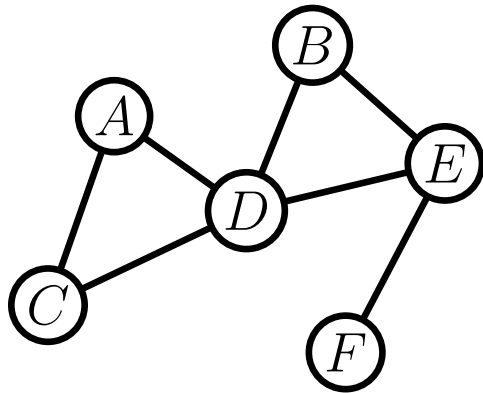


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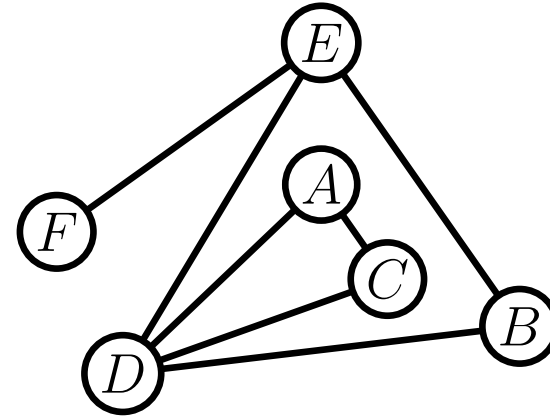
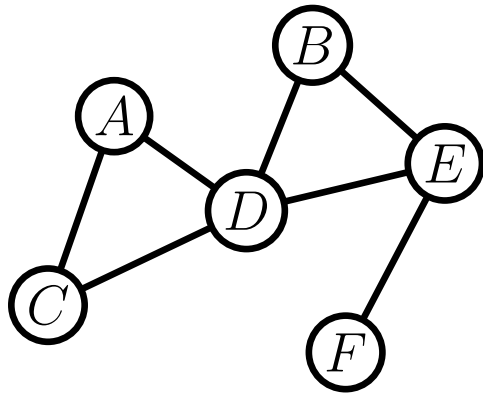
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who the people are, that is the set V of objects; and,
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The picture with circles and links is a convenient *visualization* of the graph.

Graphs and Different Types of Relationships

Affiliation graphs

Students and their courses.

Conflict graphs

Courses with students in common conflict. (Why?)

Graphs and Different Types of Relationships

Affiliation graphs

- A
- B
- C
- D
- E
- F

- 1100
- 1200
- 2200
- 2300
- 2400

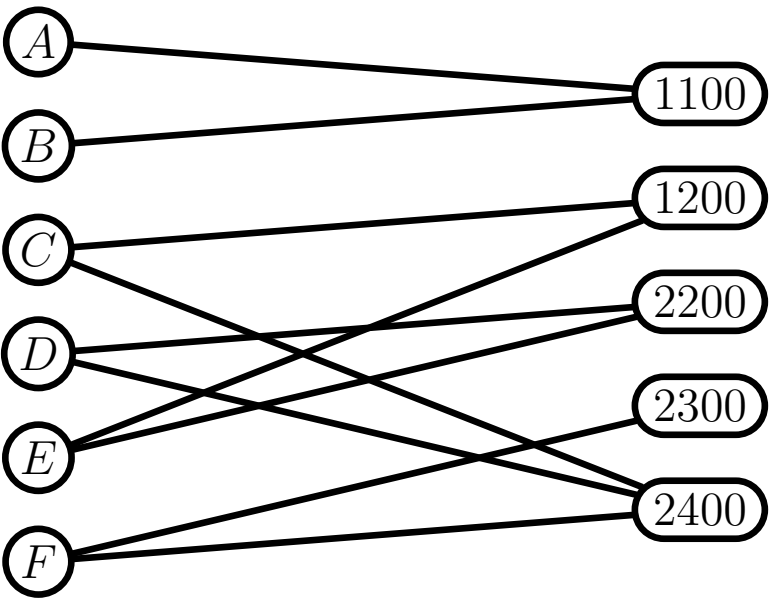
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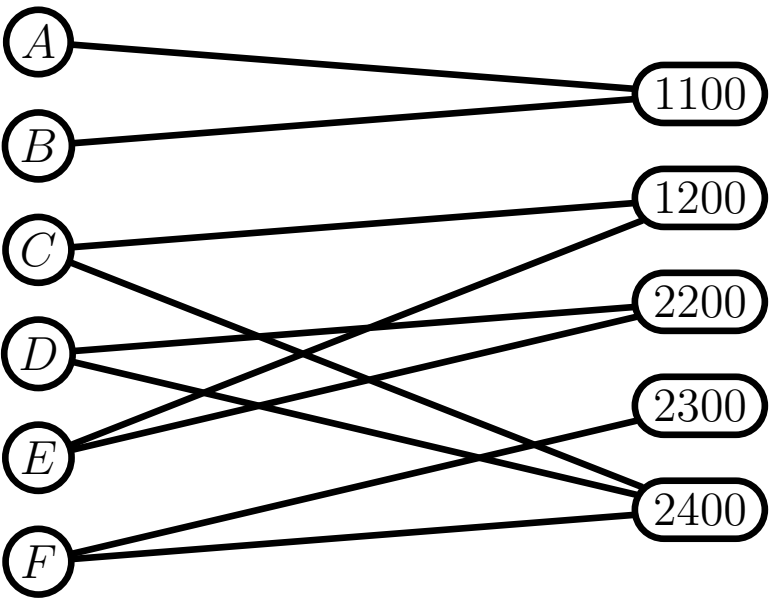
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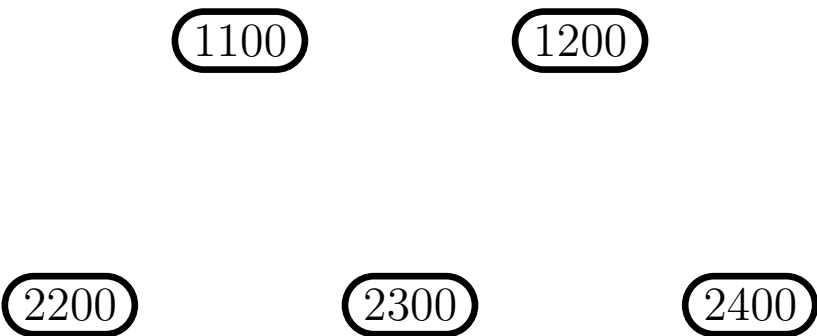
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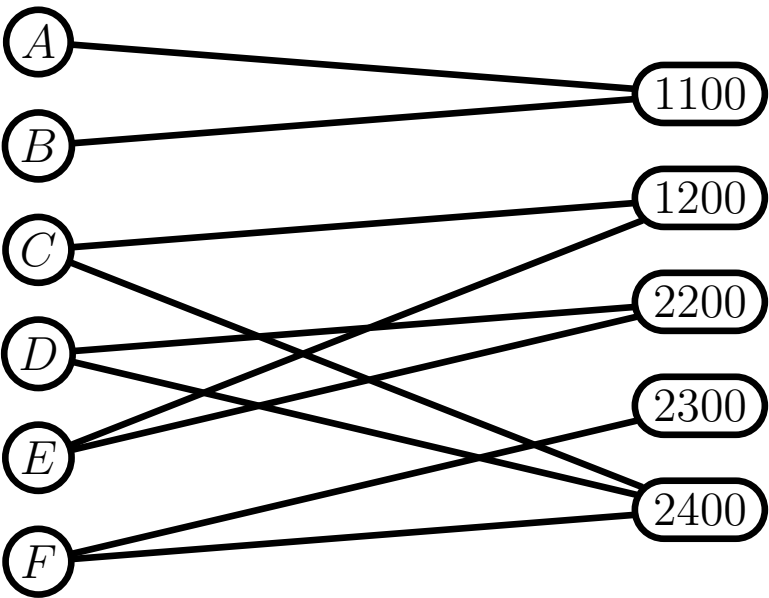
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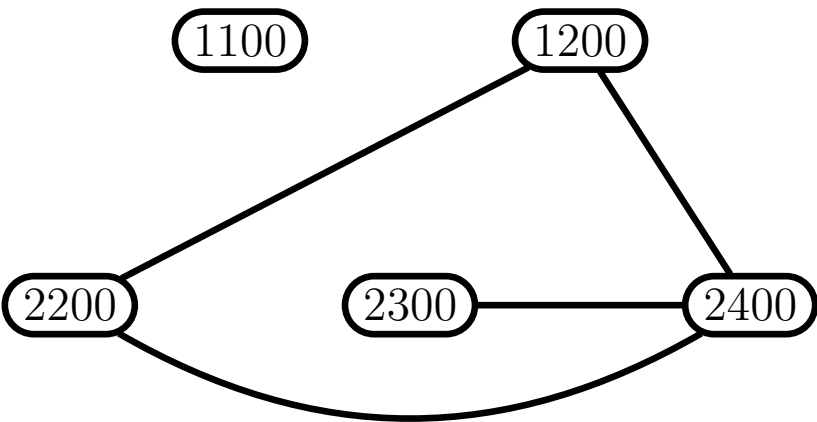
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Proof

It is Human to seek verification – proof.

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- The sun will rise tomorrow. It has risen every morning in history! (*inductive proof*)

Do you have any doubts?

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There are 4 rounds, *ergo* at most $4 \times 3 = 12$ people can be met.

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Do you have any doubts? That's the beauty of deductive proof.

When is a Number a Square

Tinker!

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n	0	± 1	± 2	± 3	± 4	± 5	± 6	± 7	± 8	± 9	± 10	± 11	...
n^2													

When is a Number a Square

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n	0	± 1	± 2	± 3	± 4	± 5	± 6	± 7	± 8	± 9	± 10	± 11	...
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Even squares come from even numbers and even numbers have even squares.

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(i) n is even

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(i) n is even $\rightarrow n = 2k \rightarrow n^2 = 2(2k^2) \rightarrow n^2$ **is even**.

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Conjecture.

Even squares come from even numbers and even numbers have even squares.

Proof. (How do I convince you this is true, *without a doubt?*) Let's look at the *cases*

- (i) n is even $\rightarrow n = 2k \rightarrow n^2 = 2(2k^2) \rightarrow n^2$ **is even**.
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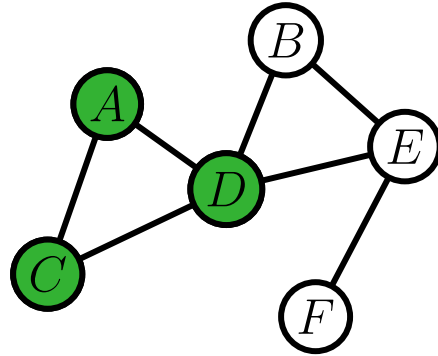
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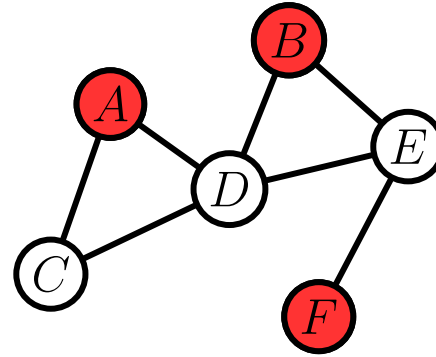
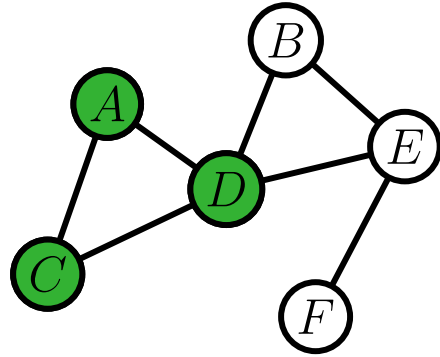
Every even square came from an even number and *every* even number has an even square.

3-war or 3-clique



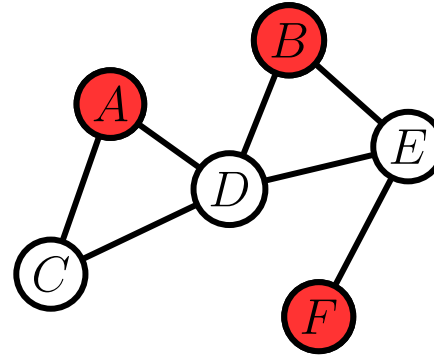
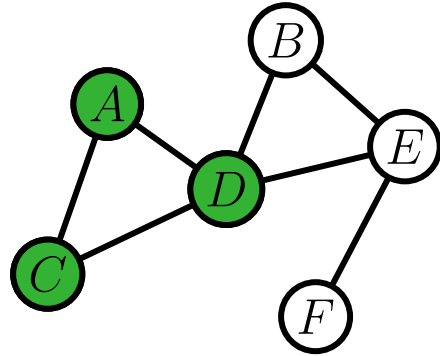
● friend clique

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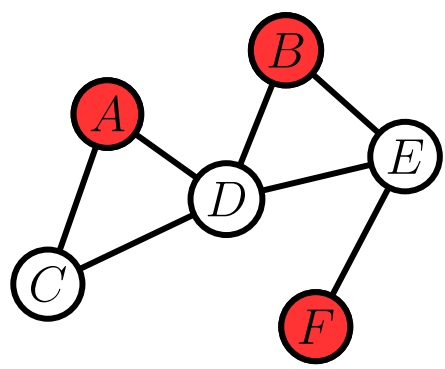
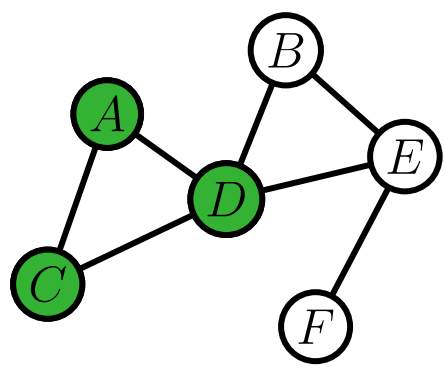


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Theorem.

Any 6-person friend network, has a 3-person friend clique or a 3-person war (or both).

3-war or 3-clique

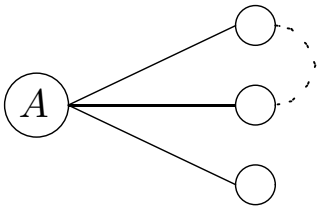


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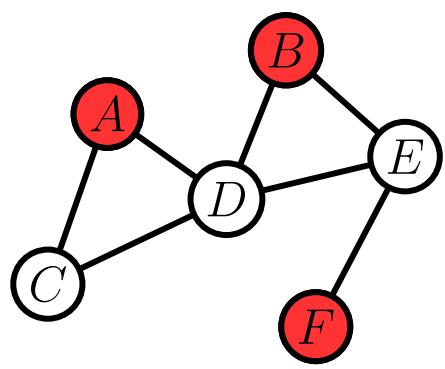
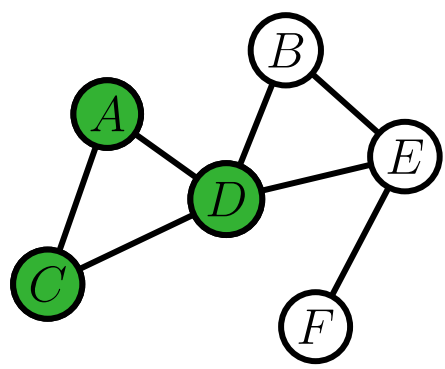
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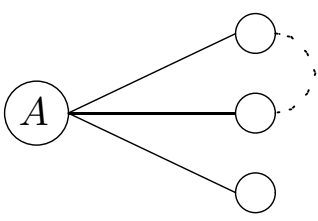


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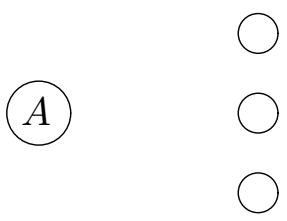
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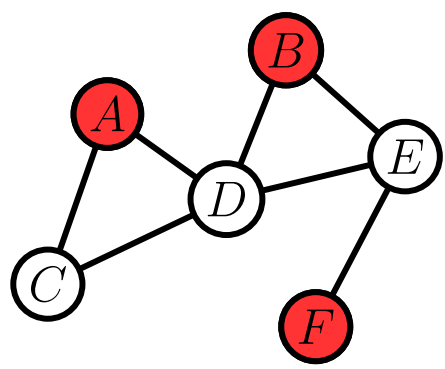
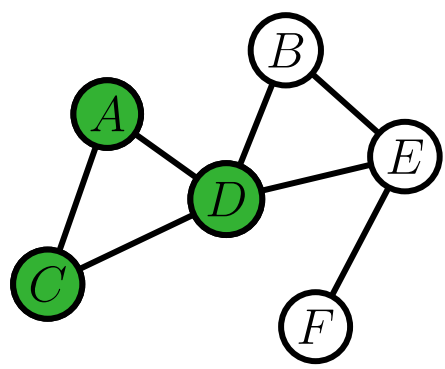
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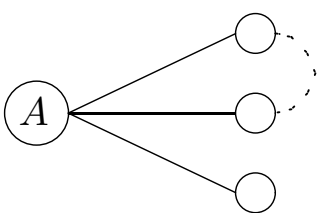


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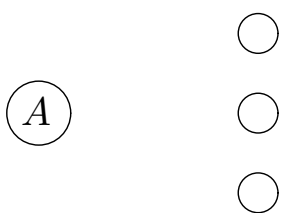
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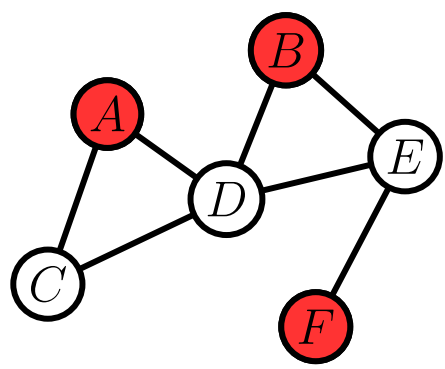
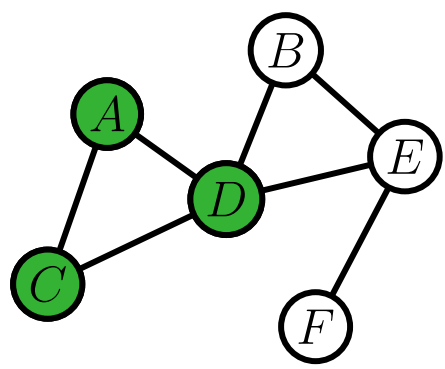
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Two friends are linked $\rightarrow$ 3-clique.



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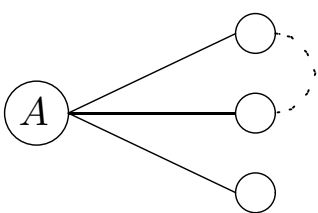


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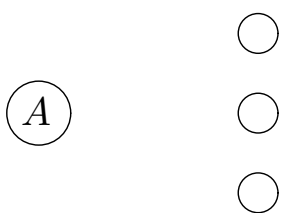
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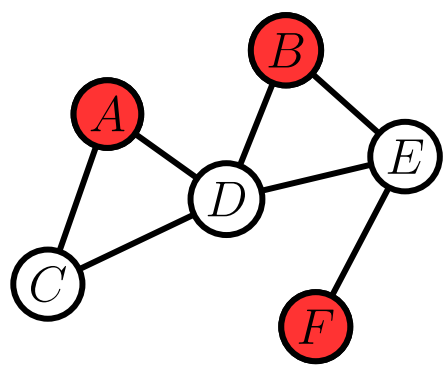
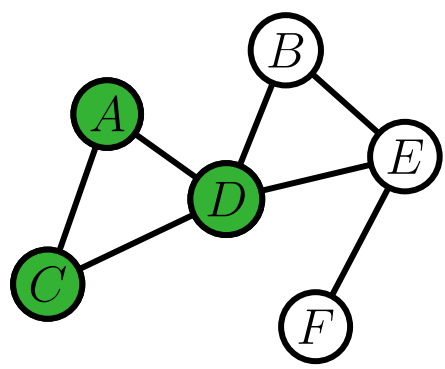
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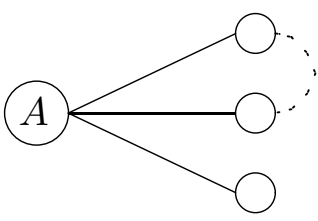


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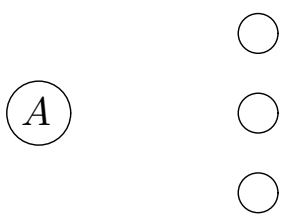
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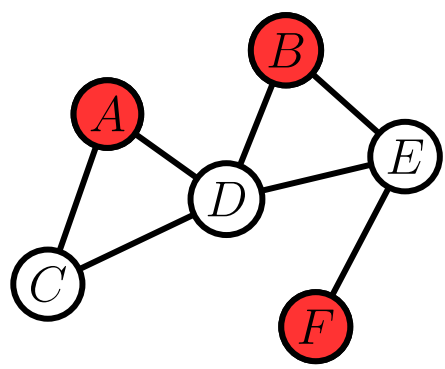
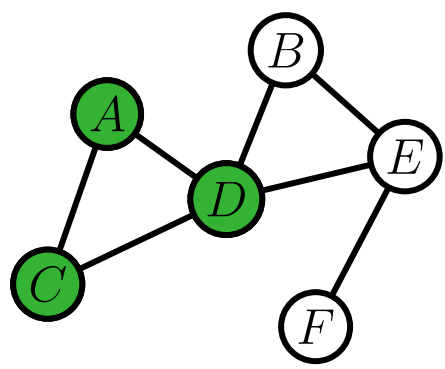
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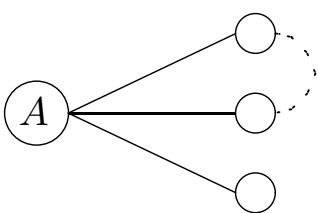


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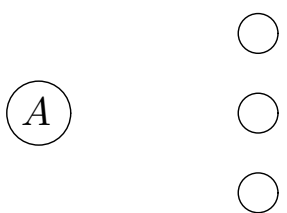
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Axiom. The Well-Ordering Principle

Any non-empty subset of $\mathbb{N} = \{1, 2, 3, \dots\}$ has a minimum element.

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Exercises.

- Construct a subset of $\mathbb{Z}$ (integers) that has no minimum element.
- Construct a positive subset of $\mathbb{Q}$ (rationals) that has no minimum element.

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$$\sqrt{2} = \frac{a_*}{b_*} \quad \rightarrow \quad a_*^2 = 2b_*^2 \quad \rightarrow \quad a_* \text{ is even (why?).}$$

So, $a_* = 2k$ and

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So, a_* **and** b_* **have the factor 2 in common.**

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It may not be so.

In which case $\sqrt{2}$ is rational,

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where $a_1, a_2, \dots$ are all integers and $b_1, b_2, \dots$ are all natural numbers.

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Next. How to make precise claims.